Session 6 : Training Your Own AI Model

You've been learning **ABOUT** AI for 5 sessions. Today, you actually **build one**. Real data. Real training. Real experiments.

Objectives:

By the end of this session, you'll be able to:

01

Know Kaggle

Understand what Kaggle is and why AI builders use it

02

Train Your AI

Train your own Cats vs Dogs AI using Teachable Machine

03

Test It Out

Test your AI on photos it has never seen before

04

Run Experiments

See what happens with 1 photo vs 10 vs 100

05

See Overfitting

Watch overfitting happen with your own eyes

Recap & Homework

Welcome back! Let's see your detective work. 🕵️

Your Cats vs Dogs Homework

- What clues (features) did you pick?
- What might trick the AI?

Quick Check from Last Session

- What's the difference between **training data** and **testing data**?
- What's **overfitting** in your own words?
- What's "**garbage in, garbage out**"?

Warm-Up: Where Do AI Builders Get Photos?

Quick question: If you wanted to train an AI to tell cats from dogs… where would you get **thousands** of cat and dog photos?

→ Take them yourself?

Too slow! You'd be there forever.

→ Search Google?

Tricky to organize and download at scale.

→ Use a ready-made website?

There IS one - and it's called **Kaggle**. 🎉

What is Kaggle? 🌐

Kaggle = a free website where AI builders share datasets with each other.

📚 Like a Library

But for AI training data instead of books

🏪 Like a Grocery Store

But stocked with datasets for AI builders

📦 Datasets for Everything

Cats, dogs, songs, sports, weather... and more

🏆 Competitions

AI builders compete to build the best models

👥 Community

A whole world of AI fans sharing knowledge

ⓘ Today, we're using a Kaggle dataset: **Cats vs Dogs!** 🐱🐶

Our Dataset - Cats vs Dogs!

Our dataset comes from:

kaggle.com - Cat and Dog Dataset

Here's what's inside the folder we'll be using:

```
cats-and-dogs-dataset/
├── training_set/    ← For TEACHING the AI
│   ├── cats/        ← LOTS of cat photos
│   └── dogs/        ← LOTS of dog photos
└── test_set/        ← For TESTING the AI
    ├── cats/        ← NEW cat photos (AI never saw!)
    └── dogs/        ← NEW dog photos (AI never saw!)
```

Training Set

This is where the AI **practices**. Like studying before an exam.

Test Set

This is the **real exam**. Photos the AI has never seen before.

Mini Activity 1: What Should Be in the Data?

Before we open Teachable Machine, let's think: If we want our AI to recognize cats and dogs **anywhere**, what kinds of photos should our dataset have?

Different Breeds

Husky, Poodle, Persian, Tabby - variety matters!

Different Colors

Brown, white, black, gray - all shades count

Different Backgrounds

Home, park, street - real-world variety

Different Lighting

Bright, dim, sunny - lighting changes everything

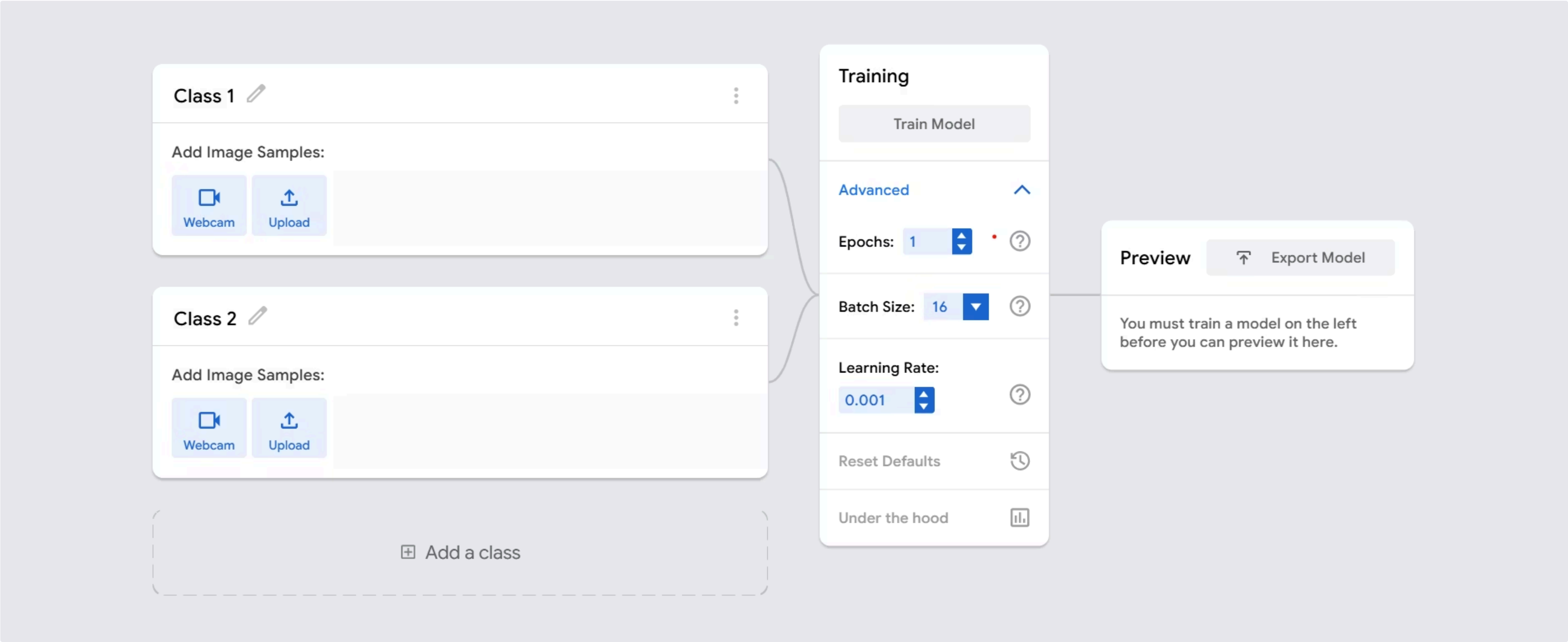
Different Angles

Front, side, looking away - all perspectives

What could go wrong if we only had photos of **white cats** and **brown dogs**? Think about it!

Time to Build: Open Teachable Machine!

We'll use **Google Teachable Machine**, a free tool that lets you train simple AI models without coding. Here's what you'll see:



Today's plan: **4 experiments**. Each one will teach us something different about how AI learns.

Experiment 1

Train with just **1 photo** of each

Experiment 2

Train with **10 photos** of each

Experiment 3

Train with **~100 photos** of each

Experiment 4

Same data, but train **way longer**

Same test photos every time. We'll see what changes! 🔬

The Test Photos Stay the Same!

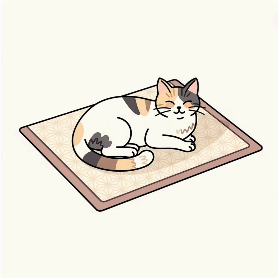
Our 4 Test Photos

We're going to test our AI 4 times. Each time, we'll use the **same 4 photos**:

These photos are brand new to the AI - it has **never seen them during training**. That's what makes them a fair test.

🐱 Cat Photo #1

A cat the AI was **never trained on** - completely unseen test data.



🐱 Cat Photo #2

A second cat photo the AI has **never encountered** before today.

🐶 Dog Photo #1

A dog the AI was **never trained on** - fresh and unseen test data.

🐶 Dog Photo #2

The final test photo - another dog the AI has **never seen** before.

Why the Same Photos Every Time?

So we can **compare**! If the AI gets better between experiments, we know it was the **data** that made the difference - not the test.

🧪 Experiment 1: Just 1 Photo Each

Prediction time! What do YOU think will happen if we only train with 1 cat photo and 1 dog photo? (Make your guess before we run it!)

Steps to Follow:

01

Upload 1 cat photo

Label it "Cat"

02

Upload 1 dog photo

Label it "Dog"

03

Click "Train Model"

Watch the AI learn!

04

Test on our 4 photos

Record the results below

Experiment 1: Results

Test the AI on our 4 photos and fill in the table:

Test Photo	Cat %	Dog %	Right?
Test cat 1	____%	____%	?
Test cat 2	____%	____%	?
Test dog 1	____%	____%	?
Test dog 2	____%	____%	?

Score: ___ out of 4 correct

👉 If training photos right, test photos wrong

It **MEMORIZED** - it didn't actually learn anything general

👉 If both right

It actually **LEARNED** - it can generalize to new photos

ⓘ Quick test: Upload one of the photos you USED for training. Does the AI get it right with super high confidence (like 99%)? Now try the test photos. Does it do as well?

🧪 Experiment 2: 10 Photos Each

Prediction: Will 10 photos each make it better?

Steps to Follow:

01

Clear the old data

Start fresh - remove Experiment 1's photos

02

Upload 10 cat photos

Label them all "Cat"

03

Upload 10 dog photos

Label them all "Dog"

04

Train again

Click "Train Model"

05

Test on the SAME 4 photos

Use the exact same test photos as before!

Experiment 2: Results

Test Photo	Cat %	Dog %	Right?
Test cat 1	____%	____%	?
Test cat 2	____%	____%	?
Test dog 1	____%	____%	?
Test dog 2	____%	____%	?

Score: ___ out of 4

Compare to Experiment 1:

Experiment	Photos	Correct
Exp 1	1 photo each	___ / 4
Exp 2	10 photos each	___ / 4

Did more data help? What do you notice?

🧪 Experiment 3: About 100 Photos Each

❓ **Prediction:** Will ~100 photos each be way better?

Steps to Follow:

01

Clear old data

Remove all photos from Experiment 2

02

Upload ~100 cat photos

Label them all "Cat"

03

Upload ~100 dog photos

Label them all "Dog"

04

Train the model

This might take a little longer!

05

Test the same 4 photos

Same test photos - always!

Experiment 3: Results

Test Photo	Cat %	Dog %	Right?
Test cat 1	____%	____%	?
Test cat 2	____%	____%	?
Test dog 1	____%	____%	?
Test dog 2	____%	____%	?

Score: ___ out of 4

Compare All Three Experiments:

Experiment	Photos	Correct
Exp 1	1 each	___ / 4
Exp 2	10 each	___ / 4
Exp 3	~100 each	___ / 4

Big jump? **More data = smarter AI!**

Mini Activity 2: What's Happening?

Quick pause to think: **Why did more photos make the AI better?**

More Photos

More variety in what the AI sees

More Variety

AI sees more kinds of cats and dogs

More Kinds

AI learns what **really** makes a cat or dog

Real Learning

AI can recognize new photos it's never seen

With just 1 photo...

The AI was **guessing**. It had almost nothing to go on.

With ~100 photos...

The AI was actually **learning** real patterns and features.

🧪 Experiment 4 - Same Data, Train LONGER

Last experiment. What if we keep the data, but **train more**?

In Teachable Machine, there's a setting called Epochs - that's how many times the AI studies ALL the photos in one round of training.

1 Epoch	50 Epochs
Going through all your flashcards **ONE time**	Going through them **50 times** - lots more practice!

⚠ **Prediction:** If we train 50 times instead of just once, will the AI get even smarter? Or... could something go WRONG?

Steps:

- Keep the same ~100 photos
- Open "Advanced" settings
- Change Epochs from 1 → 50
- Train, then test the same 4 photos

Experiment 4: Results

Test Photo	Cat %	Dog %	Right?
Test cat 1	____%	____%	?
Test cat 2	____%	____%	?
Test dog 1	____%	____%	?
Test dog 2	____%	____%	?

Experiment	Training	Correct
Exp 3	1 epoch	___ / 4
Exp 4	50 epochs	___ / 4

Better than Exp 3?

More practice helped the AI learn patterns even better!

Worse than Exp 3?

Uh oh... that might be **overfitting**! The AI memorized training photos instead of learning.

About the same?

The AI maxed out on what it could learn. More training didn't hurt OR help.

All three outcomes are real and totally normal - there's no "right" result here. The point is to **see what happens**.

The Big Picture: All 4 Experiments

Experiment	Photos	Training	Correct
🧪 Exp 1	1 each	1 epoch	___ / 4
🧪 Exp 2	10 each	1 epoch	___ / 4
🧪 Exp 3	~100 each	1 epoch	___ / 4
🧪 Exp 4	~100 each	50 epochs	___ / 4

Which experiment worked best? You just ran **real AI experiments** - just like real AI engineers do.

What Did We Learn?

Big takeaways from our experiments:

1	2
Too Little Data AI can only **memorize**, can't really learn anything general	More Data AI sees real patterns and gets **way smarter**

3	4
Way More Data Even better - up to a point!	Too Much Training Can cause **overfitting** - the AI memorizes instead of learning

ⓘ This is exactly what happens in **REAL AI systems** too - just with millions of photos instead of 100.

How Real AI Compares

The experiments you just ran? Real engineers do this too - just **bigger**:

What	You did	Real AI
Photos	~200	1 million+
Training time	A few seconds	Days to weeks
Accuracy	Pretty good	95%+

You just took your **first real step** into Machine Learning!

What We Learned Today

Kaggle

Where AI builders find real datasets

Training vs Test Set

Practice photos vs exam photos

More Data = Better AI

We proved it with our own experiments!

Too Little = Memorizing

Not enough data means no real learning

Epochs

How many times the AI studies the data

Overfitting

Too much training = memorizing, not learning

You trained AI today. That's a real skill.

Your Homework: Find a Dataset

Tonight, you're going dataset hunting! 🔍

Find one dataset online - could be **anything** you find interesting!

Places to Look:

- 🌐 **Kaggle.com** → datasets section
- 🏛️ Government websites (weather, population, sports stats)
- 🏀 Sports websites
- 🎵 Music charts

Answer These 3 Questions:

🗒 Write 1–2 sentences for each question.

01

What kind of data is in it?

Pictures? Numbers? Text? Audio?

02

What could an AI learn from it?

Example: predicting house prices

03

What might go wrong?

Could the data be unfair? Missing something?

Quick Reflection

Before we end - one question:

What was the most surprising thing you learned today?

That AI is so dependent on data?

Without good data, even the best AI is helpless

That too much training can be BAD?

Overfitting is a real danger - more isn't always better

That YOU could train an AI in minutes?

You did something real engineers do - just faster!

Something else?

Tell your mentor - and tell someone at home tonight!

Next Time: Session 7

What is Generative AI?

So far, AI has been **recognizing** things - cats, dogs, gestures. Next session, we explore AI that **CREATES** things:

✍️ Text

Stories, poems, essays - written by AI

🎨 Images

Art, drawings, illustrations - generated from words

🎵 Music

Songs, jingles, soundtracks - composed by AI

🎬 Videos

Clips, animations - created from prompts

ⓘ It's called **Generative AI** - and it's everywhere. Bring your dataset homework! 📊